Price elasticity and the cinema pricing decision

Student tasks | Original TBS lesson for A level Business (Edexcel 9BS0 content, Theme 1: Marketing and people, section 1.2.4)

The manager of Starlight Cinema wants to raise revenue without reducing contribution (revenue minus variable costs). She suggests cutting the ticket price and raising the price of the snack combo. Use price elasticity of demand (PED) to test the plan, then decide whether the evidence supports it.

Before you start you should know: PED = percentage change in quantity demanded / percentage change in price. For almost all goods PED is negative. If the size of PED is above 1 demand is elastic; below 1 it is inelastic; if it is exactly -1 demand is unit elastic. You need a calculator. This is original classroom practice, not an exam paper. Starlight Cinema and every figure are fictional.

Starter: elastic or inelastic?

Write elastic, inelastic or unit elastic for each PED value.

PED value	Elastic, inelastic or unit elastic?
(i) -2.5	
(ii) -0.3	
(iii) -1.0	
(iv) -1.2	

Task 1: tickets

Last month Starlight tested a lower ticket price for one week. Nothing else changed: the same films, the same screens and no special events.

Weekly figures (fictional)	Before	After
Ticket price	£10.00	£9.00
Tickets sold	2,000	2,300

a. Calculate the percentage change in price and in quantity demanded.

b. Calculate PED for tickets and say whether demand is elastic or inelastic.

c. Calculate weekly ticket revenue before and after, and the change.

Task 2: the snack combo

Starlight also tested a higher price for its snack combo (popcorn and a drink) for one week. Nothing else changed.

Weekly figures (fictional)	Before	After
Combo price	£6.00	£6.60
Combos sold	1,000	960

a. Calculate PED for the combo and say whether demand is elastic or inelastic.

b. Calculate weekly combo revenue before and after, and the change.

c. Explain why a business should be more willing to raise price when PED is inelastic than when it is elastic. Refer to what happens to revenue.

Task 3: revenue and contribution

Assume the variable cost is £1.50 per ticket sold and £2.50 per combo sold. Costs are simplified: treat all other costs, including film hire, as fixed and unchanged.

a. Complete the table.

Complete the table (£ per week)	Before	After	Change
Ticket revenue	20,000		
Ticket variable costs	3,000		
Ticket contribution (revenue - variable costs)	17,000		
Combo revenue	6,000		
Combo variable costs	2,500		
Combo contribution	3,500		

Working space:

b. Which gives the larger rise: ticket revenue or ticket contribution? Explain the difference in one sentence.

Task 4: why are the two products different?

Give two reasons why demand for the snack combo might be less elastic than demand for cinema tickets. Use the factors that influence PED and refer to Starlight in each answer.

Task 5: decide

The manager says: "Cut tickets and raise the combo price. Both changes will raise revenue and contribution." Write about 100 words saying how far the evidence supports her. Use at least three figures, and name one reason why each PED estimate may be unreliable.

Exit ticket

A different test raises the ticket price from £10.00 to £11.00 and tickets sold fall from 2,000 to 1,700. Nothing else changes.

a. Calculate PED.

b. What happens to ticket revenue? Show the revenue before and after.